LUNGS DISEASE PREDICTION USING MACHINE LEARNING

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MASTER OF COMPUTER APPLICATIONS

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CERTIFICATE

This is to certify that this project work entitled **"LUNGS DISEASE PREDICTION"** is the Bonafide work of **K.G.N.V.BHADRAM** who carried out the work under my supervision, and submitted in partial fulfillment of the requirements for the award of the degree, **Master of Computer Applications**, during the academic year **2024-2026.**

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The work contained in the project work is original and has been done by me under the guidance of my supervisor The work has not been submitted to any other University for the award of any degree or diploma.The guidelines of the University are followed in writing this report.

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I extend my heartfelt gratitude to the Almighty for giving me strength in proceeding with this project title **"LUNGS DISEASE PREDICTION USING MACHINE LEARNING".**

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LUNGS DISEASE PREDICTION
USING
MACHINE LEARNING

ABSTRACT

Lung diseases such as pneumonia, asthma, COPD, and lung cancer are among the leading causes of mortality worldwide. Early detection plays a crucial role in improving patient outcomes, yet traditional diagnostic methods are often time-consuming and rely heavily on clinical expertise. This project aims to develop a machine learning-based Lung Disease Prediction system using a structured CSV dataset. The proposed system preprocesses patient data, performs feature engineering, and trains a predictive model capable of identifying the likelihood of lung disease.

Various supervised learning algorithms such as Logistic Regression, Support Vector Machine (SVM), Random Forest, and Decision Tree are evaluated to determine the best-performing model. The dataset undergoes cleaning, normalization, and encoding to improve accuracy and reduce noise. The trained model is then tested on unseen data to assess its performance using metrics such as accuracy, precision, recall, and F1-score. The system ultimately provides an automated and reliable prediction that can assist healthcare professionals in early diagnosis and decision-making.

This machine learning-driven approach demonstrates the potential to enhance screening efficiency and support clinical decision systems by offering fast, accurate, and data-driven predictions. The project highlights the usefulness of predictive analytics in healthcare and provides a scalable framework for future enhancements using advanced models like deep learning and image-based classification.

Index

S.No	Topic	Page No
1	Chapter 1: Introduction	1
	1.1 Project Overview	2
	1.2 Problem Statement	2
	1.3 Project Objectives	2
	1.4 Project Scope	3
2	Chapter 2: Literature Survey	4
	2.1 Existing System Analysis	4
	2.2 Research Papers Review	4
	2.3 Similar Systems Study	4
	2.4 Technology Review	5
3	Chapter 3: System Analysis	6
	3.1 Requirement Gathering	6
	3.2 Feasibility Study	6
	3.2.1 Technical Feasibility	6
	3.2.2 Economic Feasibility	7
	3.2.3 Operational Feasibility	7
	3.3 SWOT Analysis	8
4	Chapter 4: System Requirements Specification	9
	4.1 Functional Requirements	9
	4.2 Non-Functional Requirements	10
	4.3 Hardware Requirements	10
	4.4 Software Requirements	13
	4.5 Network Requirements	15

5	Chapter 5: System Design	18
	5.1 Architecture Design	18
	5.2 UML Diagrams	18
	5.2.1 Use Case Diagrams	18
	5.2.2 Class Diagrams	19
	5.2.3 Sequence Diagrams	20
	5.2.4 Activity Diagrams	20
	5.3 Database Design	23
	5.3.1 ER Diagrams	23
	5.3.2 Data Dictionary	24
6	Chapter 6: Implementation	25
	6.1 Algorithms	25
	6.2 Pseudocode	25
	6.3 Source Code	27
	6.4 Database Queries	41
7	Chapter 7: Testing	43
	7.1 Test Plan	43
	7.2 Test Cases	43
	7.2.1 Unit Testing	44
	7.2.2 Integration Testing	44
	7.2.3 System Testing	45
	7.3 Test Results	46
8	Chapter 8: Implementation Results	49
	8.1 Screenshots	49

	8.2 Output Screens	51
	8.3 Performance Metrics	51
9	Chapter 9: Conclusion and Future Scope	52
	9.1 Conclusion	52
	9.2 Limitations	52
	9.3 Future Enhancements	52
10	References	53
11	Appendices	54
	Appendix A: User Manual	54
	Appendix B: Installation Guide	55

1.INTRODUCTION

Lung diseases are among the most critical health issues affecting millions of people worldwide. Conditions such as asthma, pneumonia, chronic obstructive pulmonary disease (COPD), and lung cancer often progress silently and may become life-threatening if not diagnosed early. Traditional diagnostic methods rely heavily on clinical tests, physician expertise, and manual evaluation, which can be time-consuming and prone to human interpretation errors. With the rapid advancement of technology, machine learning has emerged as an effective tool for analyzing medical data and improving the accuracy of disease detection.

This project focuses on developing an automated lung disease prediction system using machine learning techniques applied to a structured CSV dataset. The dataset includes various patient attributes such as age, symptoms, lifestyle habits, medical history, and lung-related measurements. By analyzing these features, machine learning algorithms can identify hidden patterns and correlations that may not be easily noticeable through conventional methods.

The main objective of this project is to build a model that can accurately classify whether a patient is at risk of developing a particular lung disease. The system performs data preprocessing, trains multiple machine learning models, evaluates performance metrics, and selects the best model for prediction. This approach not only accelerates the diagnostic process but also provides a reliable decision-support tool for healthcare professionals.

By leveraging data-driven insights, this project showcases how machine learning can contribute to early detection, reduce diagnosis time, and support better clinical outcomes for patients suffering from lung diseases

1.1 Project Overview

This project aims to develop a machine learning-based system that predicts lung diseases using a CSV dataset containing patient medical information. The data includes features such as symptoms, age, smoking habits, and medical history. After preprocessing the data, different machine learning algorithms are trained to identify disease patterns and classify the patient's lung condition. The system provides fast, accurate, and automated predictions to support early diagnosis and assist healthcare professionals.

1.2 Problem Statement

Early detection of lung diseases is challenging because traditional diagnosis methods require expert analysis, are time-consuming, and may lead to delays in treatment. With increasing patient data, manual evaluation becomes difficult and less efficient. There is a need for an automated system that can analyze patient medical information from a CSV dataset and accurately predict lung diseases to support faster and more reliable diagnosis. Project Objectives

1.3 Project Objectives

The main objective of this project is to develop an efficient machine learning-based system that can accurately predict lung diseases using a CSV dataset containing patient medical information. The project aims to preprocess and analyze key features such as symptoms, age, smoking habits, and medical history to identify meaningful patterns related to lung conditions. Multiple machine learning algorithms are trained and evaluated to select the best-performing model for disease classification. By providing a fast, automated, and reliable prediction output, the system seeks to support early diagnosis and assist healthcare professionals in making informed clinical decisions through a data-driven approach.

1.4 Project Scope:

The scope of this project is to build a machine learning-based system that predicts lung diseases using a CSV dataset containing patient medical details. It includes data preprocessing, model training, testing, and evaluating different algorithms to achieve accurate predictions. The system can take new patient input and provide quick, automated results to support early diagnosis.

2. Literature Survey

2. Literature Survey

2.1 Existing System Analysis

In the existing healthcare system, lung disease diagnosis mainly relies on traditional clinical procedures such as physical examination, medical history assessment, X-ray interpretation, and laboratory tests [1]. These methods require trained medical experts, are often time-consuming, and may delay timely diagnosis and treatment. Manual evaluation also increases chances of human error, inconsistency, and misinterpretation [2]. Furthermore, traditional approaches lack an automated mechanism for quick analysis of patient data, making early detection of lung diseases less efficient [3].

2.2 Research Papers Review

Recent research indicates that machine learning models such as Logistic Regression, Support Vector Machine (SVM), and Random Forest show high effectiveness in predicting lung diseases [4]. Many studies use structured CSV-based patient datasets and demonstrate that proper preprocessing and feature selection significantly improve prediction accuracy [5]. Overall, existing literature confirms that machine learning provides faster, automated, and more accurate lung disease prediction compared to traditional diagnostic techniques [6].

2.3 Similar Systems Study

Several healthcare prediction systems have been designed using machine learning techniques to classify diseases based on patient records [7]. Existing applications focus on predicting conditions such as heart disease, diabetes, and respiratory disorders using clinical features, symptoms, age, and lifestyle factors [8]. A few systems also attempt lung disease prediction; however, most are limited to detecting specific disorders like pneumonia or asthma and do not handle multiple lung conditions comprehensively [9].

2.4 Technology Review

The proposed project utilizes machine learning technologies to predict lung diseases from CSV-formatted patient datasets [10]. Python serves as the primary programming language, supported by libraries such as Pandas and NumPy for data processing, and Scikit-learn for model development and evaluation [11]. Classification algorithms including Logistic Regression, Random Forest, and SVM are used to build predictive models [12]. These technologies enable efficient preprocessing, feature extraction, and accurate lung disease prediction [13].

3. System Analysis

3.1. Requirement Gathering

The requirement gathering for this project involves collecting relevant patient data in CSV format, including age, symptoms, medical history, smoking habits, and other lung-related features. The system requires tools for data preprocessing, analysis, and machine learning model development. Software requirements include Python, Jupyter Notebook, and libraries like Pandas, NumPy, and Scikit-learn. The project also requires hardware capable of running data analysis and training models efficiently. The goal is to ensure sufficient and accurate data for building a reliable lung disease prediction system.

3.2 Feasibility Study

The project is technically feasible as it uses readily available machine learning libraries and tools in Python, which can handle CSV-based patient data efficiently. Economically, the project is cost-effective since it only requires standard hardware and free open-source software. Operationally, the system is practical as it can assist healthcare professionals by providing fast and accurate lung disease predictions, improving early diagnosis and decision-making. Overall, the project is feasible and implementable within the available resources and timeframe.

3.2.1 Technical Feasibility

The project is technically feasible as it uses Python and popular machine learning libraries such as Pandas, NumPy, and Scikit-learn for data processing, analysis, and model development. The CSV dataset format is easy to handle, and the system can be implemented on standard hardware without requiring specialized equipment. Jupyter Notebook provides an effective platform for coding, testing, and visualizing results, ensuring smooth development and deployment of the lung disease prediction system.

3.2.2 Economic Feasibility

The project is economically feasible as it uses open-source software such as Python, Jupyter Notebook, and machine learning libraries, which are free to use. It does not require expensive hardware or additional infrastructure, and development can be completed using standard computing resources. Overall, the project can be implemented at minimal cost while providing significant. or additional infrastructure, and development can be completed using standard computing resources. Overall, the project can be.

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3.2.3 Operational Feasibility

The project is operationally feasible because it can be easily used by healthcare professionals to input patient data and obtain lung disease predictions. The system is user-friendly, requires minimal training, and automates the diagnostic support process, reducing manual effort and improving efficiency in clinical decision-making.

3.3 SWOT Analysis

The Lung Disease Prediction Using Machine Learning project has several strengths, including fast and automated predictions, improved diagnostic accuracy, and support for healthcare professionals in decision-making. Its weaknesses include reliance on the quality and completeness of the CSV dataset and limited ability to handle real-time clinical data. Opportunities exist to expand the system with more diseases, larger datasets, or integration with hospital management systems. Threats include potential errors from biased or insufficient data, changes in medical guidelines.

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4.System Requirements Specification

4.1 Functional Requirements:

The system must allow the user to upload a CSV file containing patient health parameters required for lung disease prediction.

4.1.1 Data Input Requirements

The system must allow the user to upload a CSV file containing patient health parameters required for lung disease prediction.

4.1.2 Data Validation Requirements

The system must verify that the CSV file contains correct column names, valid data types, and no missing essential values.

4.1.3 Data Preprocessing Requirements

The system must preprocess the CSV data by handling missing values, encoding categorical data, and scaling numerical features before prediction.

4.1.4 Model Loading Requirements

The system must load a pre-trained machine learning model (e.g., Random Forest, Logistic Regression) from storage to perform predictions.

4.1.5 Prediction Requirements

The system must classify each record in the CSV file into lung disease categories such as Asthma, Lung Cancer, Pneumonia, or Normal.

4.1.6 Output Display Requirements

The system must display the predicted disease label and confidence score for each input record clearly to the user.

4.1.7 Error Handling Requirements

The system must detect and report errors such as empty CSV files, incorrect column formats, or failed predictions with meaningful messages.

4.1.8 Performance Requirements

The system must process the CSV file and generate predictions within a reasonable time, ensuring fast and efficient execution.

4.1.9 Batch Processing Requirements

The system must support multiple record predictions at once, allowing entire

CSV files to be processed in a single run.

4.2 Non-Functional Requirements:

The lung disease prediction system should process the uploaded CSV file and provide results quickly, ensuring minimal delay.

1.Performance: The lung disease prediction system should process the uploaded CSV file and provide results quickly, ensuring minimal delay.

2.Accuracy: The prediction model should deliver high accuracy in identifying lung diseases while keeping incorrect predictions (false positives/false negatives) as low as possible.

3.Usability: The interface should be simple, clean, and easy to use, allowing users to upload CSV files and view prediction results without technical difficulty.

4.Reliability: The system should function smoothly with high availability, ensuring consistent performance and minimal downtime during predictions.

5. Security: The system must ensure secure handling of patient health data by using safe storage practices and secure communication protocols to protectsensitive information.

6.Maintainability: The system should be easy to update, allowing developers to replace or improve the prediction model and add new features.

4.3 Hardware Requirements

For lung disease prediction using CSV data, a powerful multi-core processor is recommended to efficiently handle data preprocessing and model inference.

1.Processing Power (CPU/GPU)

Central Processing Unit (CPU):

For lung disease prediction using CSV data, a powerful multi-core processor is recommended to efficiently handle data preprocessing and model inference.

- Minimum: Intel i3 / AMD Ryzen 3
- Recommended: Intel i5/i7 or AMD Ryzen 5/7 for faster computation

- Number of Cores:
    - Minimum: 4 cores
    - Recommended: 8–12 cores for faster batch processing and smoother operation

Graphics Processing Unit (GPU) (Optional – for Deep Learning Models):

If the system uses deep learning models (e.g., neural networks), a GPU significantly accelerates training.

- Recommended GPU: NVIDIA GTX/RTX series or Tesla

2. Memory (RAM)

Adequate RAM is essential for loading datasets, training models, and performing batch predictions.

- Minimum RAM: 8 GB for basic prediction tasks
- Recommended RAM: 16–32 GB for handling large CSV datasets and faster in-memory computation

Storage

Solid State Drive (SSD):

Fast storage is needed for reading/writing medical CSV data, loading models, and saving logs.

- **Minimum: 256 GB SSD**
- **Recommended: 512 GB – 1 TB SSD**, especially when storing datasets and trained models

Backup Storage:

- Cloud platforms like **Google Drive, AWS S3, or OneDrive** can be used for storing datasets, backups, and model versions.

4. Network Requirements

A stable network ensures smooth data transfers, updates, and dependency installations.

- **Minimum:** Standard broadband connection
- **Recommended:** High-speed internet (≥ 100 Mbps) for uploading/downloading datasets and accessing cloud services

5. Server or Cloud Infrastructure

On-Premise Setup:

Suitable for institutions handling sensitive medical data.

- High-performance machine with multi-core CPU, large RAM, and SSD storage
- Examples: Dell PowerEdge, HP ProLiant servers

Cloud Infrastructure:

Cloud platforms offer high scalability for model training and deployment.

- **Recommended Platforms:** AWS, Google Cloud, Microsoft Azure
- **Services Used:**
    - **Compute:** AWS EC2, Google Compute Engine
    - **Databases:** Cloud SQL, AWS RDS for storing patient records
    - **Storage:** AWS S3, Google Cloud Storage for large dataset handling

6. Power Supply & Cooling

- **UPS (Uninterrupted Power Supply):** Ensures the system runs without interruption during power failures
- **Adequate Cooling:** Prevents overheating during long model training or batch prediction tasks
- Cloud platforms like **Google Drive, AWS S3, or OneDrive** can be used for storing datasets, backups, and model versions.
- **UPS (Uninterrupted Power Supply):** Ensures the system runs without interruption during power failures

4.4 Software Requirements

1.Programming Languages

Python:

Python is the primary language used for implementing lung disease prediction models because of its simplicity and strong machine learning ecosystem.

- **Why Python?**
  - Excellent support for data preprocessing, model building, and evaluation
  - Large community and extensive documentation
- **Popular Python Libraries Used:**
  - **pandas:** For CSV data loading, cleaning, and manipu.
  - **numpy:** For numerical operations and array handling
  - **scikit-learn:** For building ML models like Logistic Regression, Random Forest, SVM, etc.
  - **TensorFlow / Keras (Optional):** For deep learning models if neural networks are used
  - **XGBoost / LightGBM:** For high-performance gradient boosting models
  - **Matplotlib / Seaborn:** For visualizing medical data, graphs, and patterns

2. Machine Learning Frameworks

scikit-learn:

A powerful library preferred for classic machine learning Training ML models (Logistic Regression, Decision Trees, SVM, Random Forest, etc.)

- Splitting data into train/test sets
- Evaluating performance (accuracy, precision, recall, confusion matrix)

TensorFlow / Keras (Optional):

Used when the project includes deep learning or neural network models.

- Suitable for building deep neural networks
- Helps in modeling complex patterns in medical datasets
- Keras provides easy API usage, while TensorFlow offers more flexibility and scalability

3. Data Storage and Databases

CSV Files (Primary Storage):

Since your project uses CSV datasets, files are stored locally and processed directly using Python.

Relational Databases (Optional):

Can be used if you want to store patient records, prediction Examples:

- MySQL
- PostgreSQL
- SQLite
- MS SQL Server

These databases help with:

- Storing structured health data
- Managing multiple patient entries
- Retrieving past prediction results

4.Development Tools and Environments

- **Jupyter Notebook / JupyterLab:** For model development, testing, and visualization
- **PyCharm / VS Code:** For coding the full project and backend system
- **Anaconda Distribution:** For managing Python libraries and environments
- **Git / GitHub:** For version control and project backup

5. Web Framework (Optional)

If the project includes a web interface:

- **Flask:** Lightweight framework for building prediction APIs and web UI

- **Streamlit:** Quick tool to build interactive ML dashboards without heavy coding

6. Operating System Support

- Windows 10/11
- Linux (Ubuntu)
- macOS

4.5 Network Requirements

Internet Bandwidth

Minimum Bandwidth:

- **10 Mbps** is sufficient for small-scale lung disease prediction systems, especially when working in a development environment or uploading small CSV files.

Recommended Bandwidth:

- **100 Mbps – 1 Gbps** is recommended for larger datasets, cloud-based model execution, or when multiple users access the system simultaneously.
- Higher bandwidth ensures smooth downloading of medical datasets, Python libraries, and fast communication with cloud platforms.

2. Low Latency

Fast Data Processing:

- Although lung disease prediction is not real-time like fraud detection, low latency helps in:
    - Faster uploading of CSV files
    - Quickly sending data to API-based prediction services
    - Returning prediction results without noticeable delay

Recommended Latency:

- **Below 200 ms** is ideal for interactive applications and smooth user experience.

3. Data Transfer & Throughput

Data Throughput:

Lung disease prediction systems may handle large CSV files with many patient records.

- The network should support efficient transfer of files ranging from a few MBs up to several hundred MBs.
- For research or hospital-level usage, the system should support higher throughput to process thousands of records at once.
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4.Secure Communication

Encrypted Data Transmission:

Patient health data (PHI) is sensitive and must be protected during transfer.

- Use **HTTPS** for secure API communication
- Enable **SSL/TLS encryption** when accessing backend servers
- Hospitals or research centers may use:
    - **VPNs (Virtual Private Networks)**
    - **Private internal networks**

    These methods ensure complete protection of medical data and compliance with health regulations.

5. Cloud Connectivity

Cloud-based Processing:

If your lung disease prediction system uses cloud services for model hosting or data storage, strong network connectivity is important.

- Cloud platforms like **AWS, Google Cloud, or Azure** may host:
  - Trained ML models
  - Medical datasets
  - Prediction APIs
  - Databases

Requirement:

- Fast and stable internet ensures smooth communication between the local system and cloud resources
- Upload/download of large CSV datasets becomes faster and more reliable

Using cloud services like AWS Direct Connect or Google Cloud Interconnect can help establish a dedicated connection for improved performance and security.s

5.System Design

5.1 Architecture Design

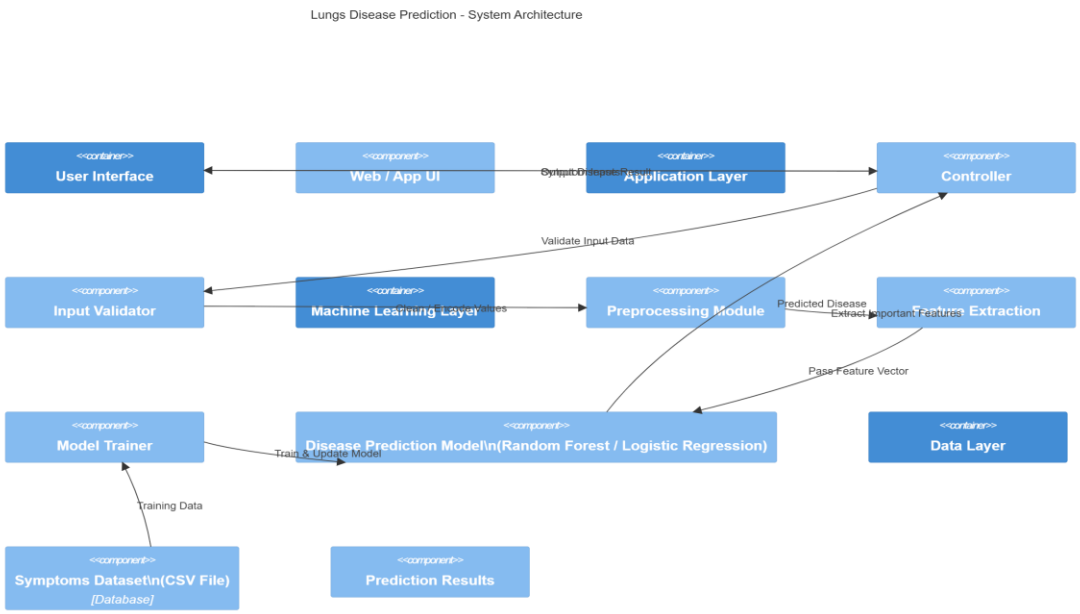

Fig1: Architecture Design

Description:

The system follows a simple machine-learning architecture. The flow begins with dataset input, followed by preprocessing, feature selection, model training, prediction, and result visualization. The architecture ensures modularity, allowing each component—data cleaning, model training, and prediction—to function independently for easy maintenance.

5.2 UML Diagrams

5.2.1 Use Case Diagram

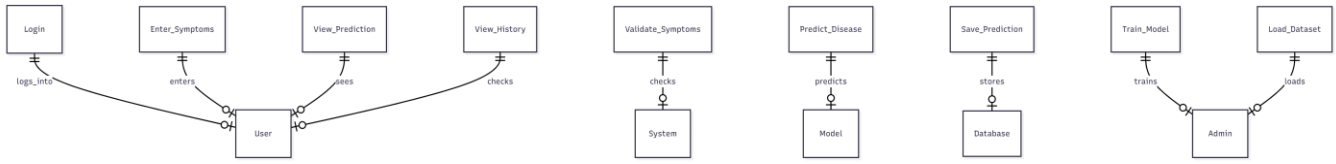

Fig2: Use Case Diagram

Description:

The use case diagram represents how a user interacts with the system. The main actor is the User, who performs actions such as uploading the dataset, training the model, entering patient details, and viewing predictions. The system responds with results and performance metrics.

5.2.2 Class Diagram

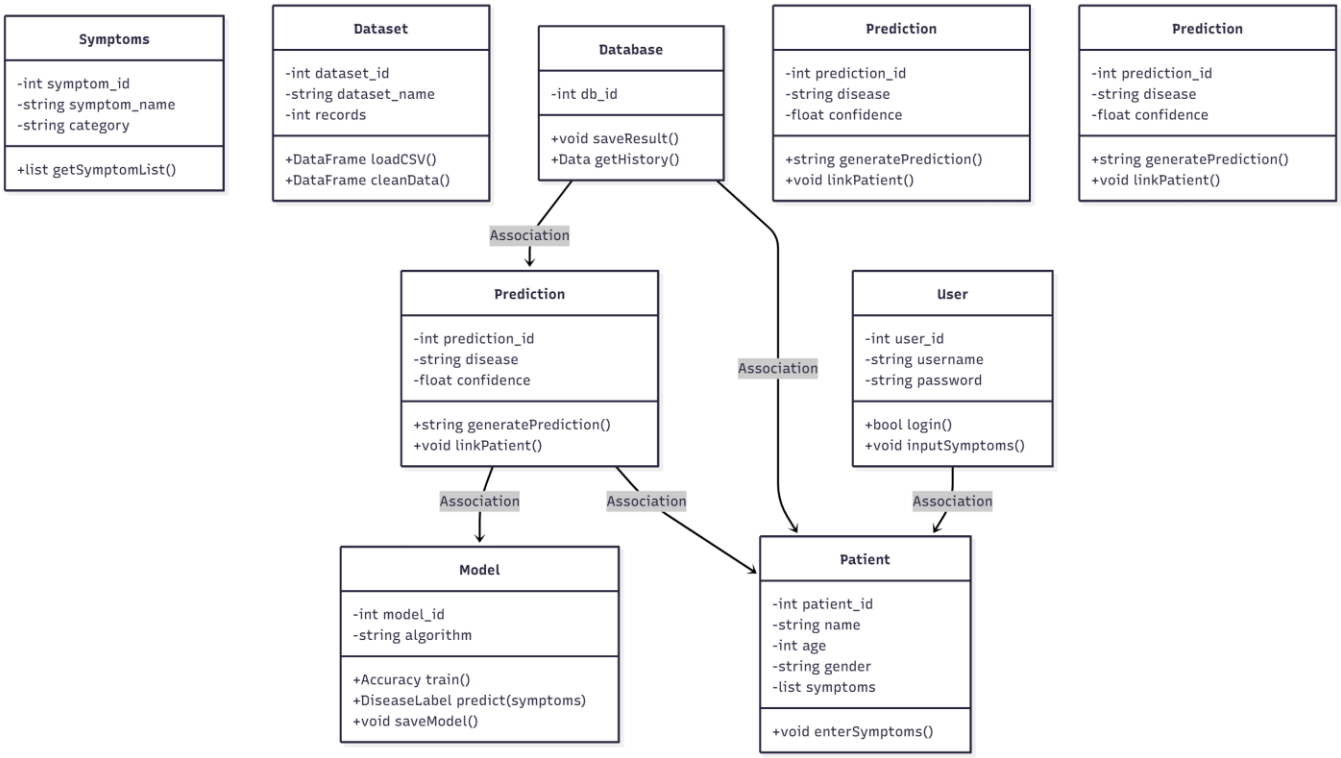

Fig3: Class Diagram

Description:

The class diagram shows the system structure in terms of classes and their relationships.
Key classes include:

- DatasetLoader – loads dataset
- Preprocessor– cleans and transforms data
- MLModel – trains and evaluates model
- Predictor – generates predictions
- Visualizer – displays charts and graphs

These classes interact to complete the prediction workflow.

5.2.3 Sequence Diagram

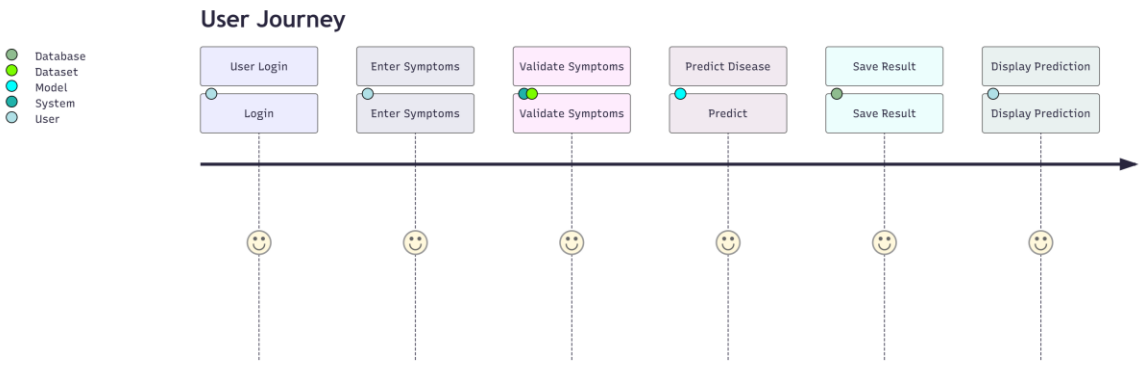

Fig 4: Sequence Diagram

Description:

The sequence diagram shows how the system processes a user request. Steps include:

1. User enters patient data
2. Data passes through preprocessing
3. ML model generates prediction
4. Result is returned to the user

This illustrates the real-time flow of actions in the system.

5.2.4 Activity Diagram

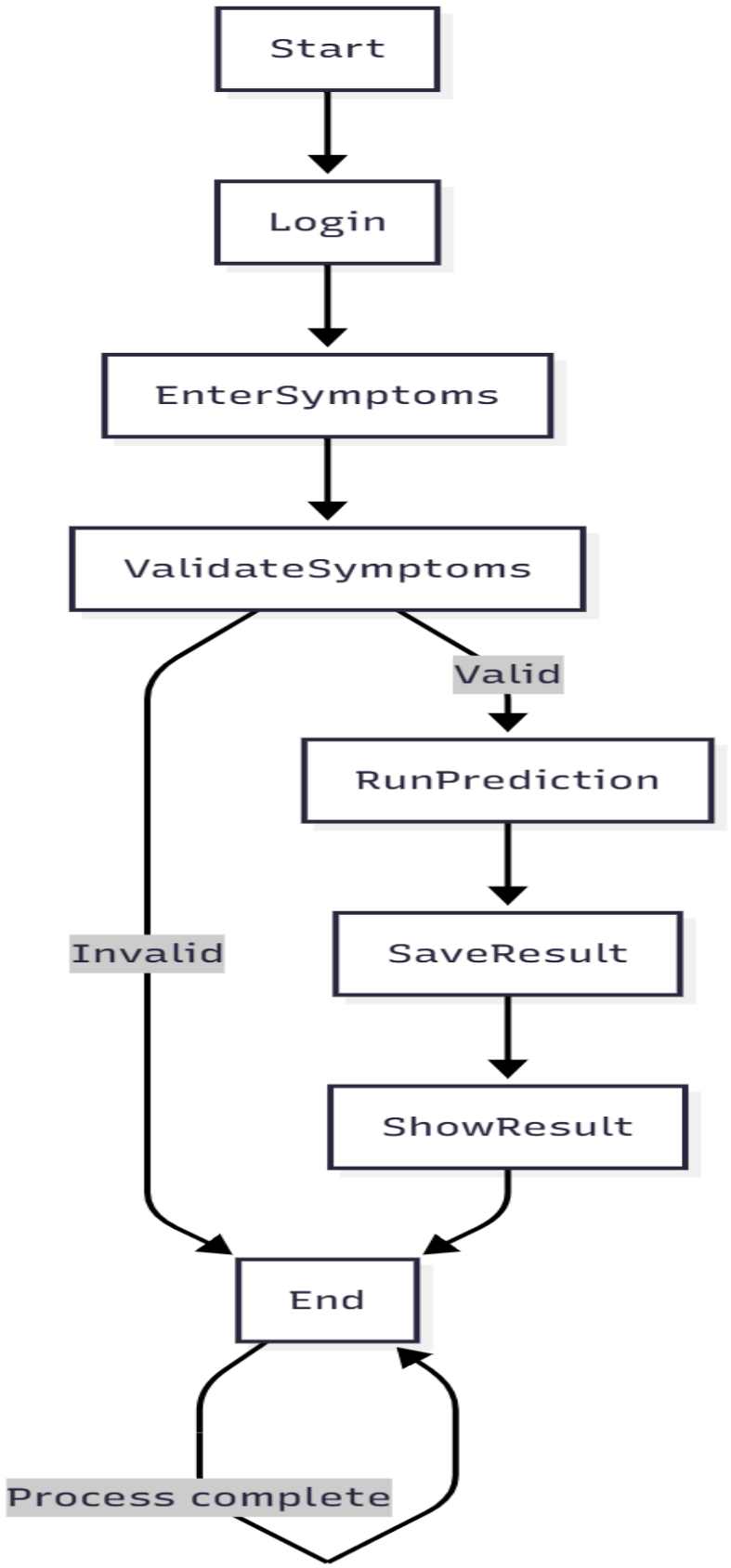

Fig 5: Activity Diagram

Description:

This diagram outlines the system's operational workflow. It includes the activities:

- Load dataset
- Preprocess data
- Train model
- Input patient details
- Generate prediction
- Display output

The flow moves from start to end in a simple, linear manner.

5.3 Database Design

5.3.1 ER Diagram

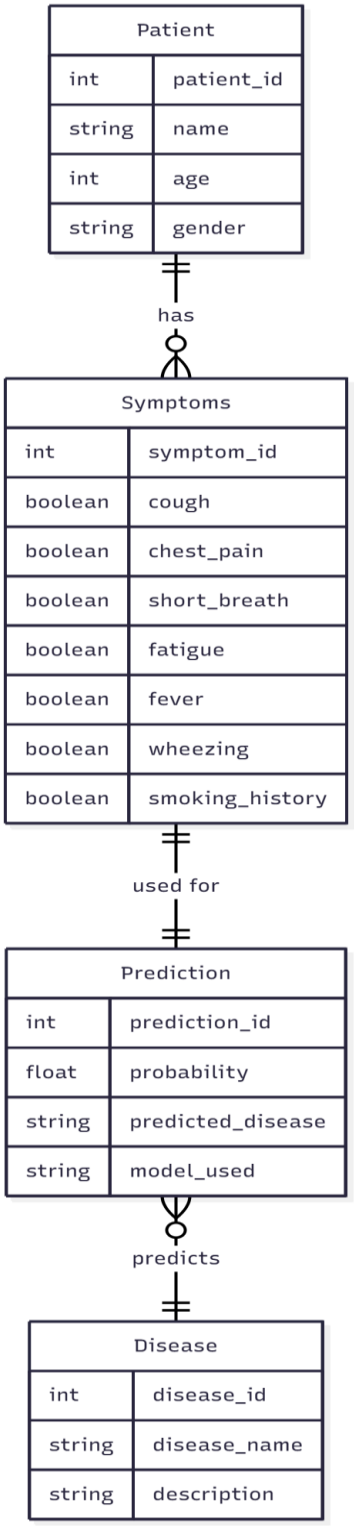

Fig 6: ER Diagram

Description:

The ER diagram contains entities such as Patient, Prediction, and User.

Relationships include:

- A user can generate multiple predictions
- Each prediction is linked to a patient record

This helps organize prediction history and patient data

5.3.2 Data Dictionary

The data dictionary lists all dataset fields along with their meanings.

Examples:

- Age – Patient age in years
- SmokingHistory – Number of years smoked
- CoughSeverity – Symptom level (0–5)
- LungCapacity – Spirometry value
- Disease – Target label (0 = No disease, 1 = Disease)

These descriptions ensure clarity in dataset usage

Database name: my sql **Table name**: User
Table 1: User

Sno	**Attribute Name**	**Data Type**	**Description**
1	User _id	INT (Primary Key)	Unique identifier for each user.
2	User_ name	VARCHAR(100)	Name of the user.
3	Email	VARCHAR(100)	Email address of the user
4	password	VARCHAR(25)	Encrypted password for user authentication
5	Registration_ data	DATETIME	Date and time of user registration

6. IMPLEMENTATION

6.1 Algorithm: Random Forest

Random Forest is an ensemble machine learning algorithm that builds multiple decision trees and combines their outputs. It improves prediction accuracy by reducing overfitting and increasing stability.

1.Load Dataset :

The lung disease dataset in CSV format is loaded into the system using data handling libraries like Pandas. This dataset contains symptoms and corresponding disease labels that will be used for training the model. Loading the dataset is the first step before preprocessing and model building.

2. DATA COLLECTION:

The data is collected from medical symptom records, publicly available datasets, or manually prepared CSV files. It includes various symptoms such as cough, fever, chest pain, and shortness of breath. This collected data forms the foundation for training the lung disease prediction model.

3. DATA PREPROCESSING:

Data preprocessing involves cleaning the dataset by handling missing values, removing noise, and converting categorical data into numerical form. The symptoms and disease labels are encoded for machine learning. This step ensures the data is ready for accurate model training.

6.2 Pseudo Code:

START

1. Load the required libraries

   - pandas

   - numpy

   - sklearn (train_test_split, classifier models, accuracy_score)

2. Load the dataset

INPUT: lungs_disease.csv

DATA ← read CSV file

3. Preprocess the dataset

- Handle missing values (if any)
- Convert categorical values to numerical (Label Encoding)
- Separate FEATURES and TARGET

FEATURES ← symptom columns

TARGET ← disease column

4. Split the dataset into training and testing sets

TRAIN_X, TEST_X, TRAIN_Y, TEST_Y ← train_test_split(DATA)

5. Select machine learning model

MODEL ← RandomForestClassifier()

6. Train the model

MODEL.fit(TRAIN_X, TRAIN_Y)

7. Predict using the test set

PRED_Y ← MODEL.predict(TEST_X)

8. Evaluate the model

ACCURACY ← accuracy_score(TEST_Y, PRED_Y)

OUTPUT "Model Accuracy:", ACCURACY

9. Take user input symptoms

- cough?
- chest pain?
- shortness of breath?
- fever?
- wheezing?
- fatigue?

USER_INPUT ← convert Yes/No to 1/0

10. Predict disease for the user

RESULT ← MODEL.predict(USER_INPUT)

11. Display predicted disease

OUTPUT "Predicted Lung Disease:", RESULT

END

6.3 Source Code:

```
"""
Lung Disease Prediction System
A complete web application with user management and ML prediction
"""
from flask import Flask, render_template, request, redirect, url_for, session, flash, jsonify
from werkzeug.security import generate_password_hash, check_password_hash
from functools import wraps
import pandas as pd
import joblib
import os
from datetime import datetime
import json

app = Flask(__name__)
app.secret_key = 'your-secret-key-change-this-in-production-12345'
app.config['UPLOAD_FOLDER'] = 'data'

# In-memory databases (for simplicity - use SQLite/PostgreSQL in production)
users_db = {
    'admin': {
        'password': generate_password_hash('admin123'),
        'name': 'Admin User',
        'email': 'admin@lungdisease.com',
```

```
        'role': 'admin',
        'created_at': datetime.now().strftime('%Y-%m-%d %H:%M:%S')
    }
}

predictions_db = {}  # Store predictions by user_id

# Load ML model
model = None
try:
    model = joblib.load('models/lung_disease_model.pkl')
    print("✓ Model loaded successfully")
except:
    print("⚠ Model not found - will be trained on first prediction")

# Decorators
def login_required(f):
    @wraps(f)
    def decorated_function(*args, **kwargs):
        if 'user_id' not in session:
            flash('Please login to access this page', 'warning')
            return redirect(url_for('login'))
        return f(*args, **kwargs)
```

```
    return decorated_function


def admin_required(f):
    @wraps(f)
    def decorated_function(*args, **kwargs):
        if 'user_id' not in session:
            flash('Please login to access this page', 'warning')
            return redirect(url_for('login'))
        user = users_db.get(session['user_id'])
        if not user or user.get('role') != 'admin':
            flash('Admin access required', 'danger')
            return redirect(url_for('dashboard'))
        return f(*args, **kwargs)
    return decorated_function


# Routes
@app.route('/')
def index():
    if 'user_id' in session:
        return redirect(url_for('dashboard'))
    return redirect(url_for('login'))


@app.route('/register', methods=['GET', 'POST'])
```

```
def register():
    if request.method == 'POST':
        username = request.form.get('username', '').strip()
        password = request.form.get('password', '')
        name = request.form.get('name', '').strip()
        email = request.form.get('email', '').strip()

        # Validation
        if not username or not password or not name or not email:
            flash('All fields are required', 'danger')
            return render_template('register.html')

        if username in users_db:
            flash('Username already exists', 'danger')
            return render_template('register.html')

        # Create new user
        users_db[username] = {
            'password': generate_password_hash(password),
            'name': name,
            'email': email,
            'role': 'user',
            'created_at': datetime.now().strftime('%Y-%m-%d %H:%M:%S')
```

```
        }

        flash('Registration successful! Please login.', 'success')
        return redirect(url_for('login'))

    return render_template('register.html')

@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        username = request.form.get('username', '').strip()
        password = request.form.get('password', '')

        user = users_db.get(username)

        if user and check_password_hash(user['password'], password):
            session['user_id'] = username
            session['user_name'] = user['name']
            session['user_role'] = user['role']

            flash(f'Welcome back, {user["name"]}!', 'success')

            if user['role'] == 'admin':
```

```
            return redirect(url_for('admin_dashboard'))
        else:
            return redirect(url_for('dashboard'))
    else:
        flash('Invalid username or password', 'danger')

    return render_template('login.html')

@app.route('/logout')
def logout():
    session.clear()
    flash('You have been logged out', 'info')
    return redirect(url_for('login'))

@app.route('/dashboard')
@login_required
def dashboard():
    user_id = session['user_id']
    user_predictions = predictions_db.get(user_id, [])
    return render_template('dashboard.html',
                           predictions=user_predictions,
                           total_predictions=len(user_predictions))
```

```
@app.route('/predict', methods=['GET', 'POST'])
@login_required
def predict():
    if request.method == 'POST':
        try:
            # Get form data
            data = {
                'age': int(request.form.get('age')),
                'gender': request.form.get('gender'),
                'smoking_history': request.form.get('smoking_history'),
                'cough': request.form.get('cough'),
                'shortness_of_breath': request.form.get('shortness_of_breath'),
                'chest_pain': request.form.get('chest_pain'),
                'fever': request.form.get('fever'),
                'wheezing': request.form.get('wheezing'),
                'spO2': int(request.form.get('spO2'))
            }

            # Train model if not loaded
            global model
            if model is None:
                from train_model import train_and_save_model
                model = train_and_save_model()
```

```
# Prepare data for prediction
df = pd.DataFrame([data])

# Encode categorical variables
df['gender'] = df['gender'].map({'M': 1, 'F': 0})
df['smoking_history'] = df['smoking_history'].map({'Yes': 1, 'No': 0})
df['cough'] = df['cough'].map({'Yes': 1, 'No': 0})
df['shortness_of_breath'] = df['shortness_of_breath'].map({'Yes': 1, 'No': 0})
df['chest_pain'] = df['chest_pain'].map({'Yes': 1, 'No': 0})
df['fever'] = df['fever'].map({'Yes': 1, 'No': 0})
df['wheezing'] = df['wheezing'].map({'Yes': 1, 'No': 0})

# Make prediction
prediction = model.predict(df)[0]
probability = model.predict_proba(df)[0]
confidence = max(probability) * 100

# Store prediction
user_id = session['user_id']
if user_id not in predictions_db:
    predictions_db[user_id] = []
```

```
            prediction_record = {
                'id': len(predictions_db[user_id]) + 1,
                'timestamp': datetime.now().strftime('%Y-%m-%d %H:%M:%S'),
                'input_data': data,
                'prediction': prediction,
                'confidence': round(confidence, 2)
            }

            predictions_db[user_id].append(prediction_record)

            return render_template('result.html',
                                   prediction=prediction,
                                   confidence=round(confidence, 2),
                                   input_data=data)

        except Exception as e:
            flash(f'Error making prediction: {str(e)}', 'danger')
            return redirect(url_for('predict'))

    return render_template('predict.html')

@app.route('/history')
@login_required
```

```
def history():
    user_id = session['user_id']
    user_predictions = predictions_db.get(user_id, [])
    return render_template('history.html', predictions=user_predictions)


@app.route('/admin')
@admin_required
def admin_dashboard():
    total_users = len([u for u in users_db.values() if u['role'] == 'user'])
    total_predictions = sum(len(preds) for preds in predictions_db.values())


    return render_template('admin_dashboard.html',
                           users=users_db,
                           total_users=total_users,
                           total_predictions=total_predictions)


@app.route('/admin/users')
@admin_required
def admin_users():
    return render_template('admin_users.html', users=users_db)


@app.route('/admin/user/delete/<username>', methods=['POST'])
@admin_required
```

```
def admin_delete_user(username):
    if username == 'admin':
        flash('Cannot delete admin user', 'danger')
    elif username in users_db:
        del users_db[username]
        if username in predictions_db:
            del predictions_db[username]
        flash(f'User {username} deleted successfully', 'success')
    else:
        flash('User not found', 'danger')

    return redirect(url_for('admin_users'))

@app.route('/admin/user/edit/<username>', methods=['GET', 'POST'])
@admin_required
def admin_edit_user(username):
    if username not in users_db:
        flash('User not found', 'danger')
        return redirect(url_for('admin_users'))

    if request.method == 'POST':
        user = users_db[username]
        user['name'] = request.form.get('name', user['name'])
```

```
        user['email'] = request.form.get('email', user['email'])

        new_password = request.form.get('new_password', '')
        if new_password:
            user['password'] = generate_password_hash(new_password)

        flash(f'User {username} updated successfully', 'success')
        return redirect(url_for('admin_users'))

    return render_template('admin_edit_user.html',
                           username=username,
                           user=users_db[username])

@app.route('/api/check-password-strength', methods=['POST'])
def check_password_strength():
    password = request.json.get('password', '')

    score = 0
    feedback = []

    if len(password) >= 8:
        score += 1
    else:
```

```
    feedback.append('At least 8 characters')

if any(c.isupper() for c in password):
    score += 1
else:
    feedback.append('One uppercase letter')

if any(c.islower() for c in password):
    score += 1
else:
    feedback.append('One lowercase letter')

if any(c.isdigit() for c in password):
    score += 1
else:
    feedback.append('One number')

if any(c in '!@#$%^&*()_+-=[]{}|;:,.<>?' for c in password):
    score += 1
else:
    feedback.append('One special character')

strength = 'Very Weak'
```

```
    if score >= 5:
        strength = 'Strong'
    elif score >= 4:
        strength = 'Good'
    elif score >= 3:
        strength = 'Medium'
    elif score >= 2:
        strength = 'Weak'

    return jsonify({
        'strength': strength,
        'score': score,
        'feedback': feedback
    })

if __name__ == '__main__':
    app.run(debug=True, host='0.0.0.0', port=5000)
```

6.4 DataBase Quarries:

6.4 Database Queries (SQLite)

```
import mysql.connector
# Function to connect to the MySQL database
```

```
# Function to make database connection

def project_db():
    try:
        # Establish the database connection
        connection = mysql.connector.connect(
            host="localhost",       # Host where MySQL server is running
            user="root",            # MySQL username
            password="",            # MySQL password (empty by default)
            database="project_db"  # Replace with your database name
        )

        if connection.is_connected():
            print("Database connection successful!")
            return connection

    except mysql.connector.Error as err:
        print(f"Error: {err}")
        return None

# Test the database connection
if __name__ == "__main__":
    db = project_db()
    if db:
        print("Closing the database connection.")
        db.close()
```

7. Testing

7.1 Test Plan

The testing phase ensures that the system behaves as expected under different scenarios. The test plan covers:

- Unit testing of data preprocessing functions
- Model training verification
- Validation of GUI/web inputs
- System testing for performance and accuracy
- Handling invalid or missing input scenarios

Testing also ensures that the prediction interface correctly processes patient values and returns appropriate disease risk.

7.2 Test Cases

7.2.1 Unit Testing

Data Preprocessing Functions:

- Test data imputation methods (e.g., filling missing values) to ensure they handle missing data correctly.
- Test encoding functions (e.g., OneHotEncoding) to verify categorical variables are transformed into the correct format.

Model Training and Prediction:

- Test individual model methods like training (fit()) and prediction (predict()) to ensure the model is learning and making predictions accurately.

Evaluation Metrics Calculation:

Test functions that compute accuracy, precision, recall, F1-score, and AUC to ensure correct implementation. The training data refers to the portion of the CKD dataset used to train the Random Forest model. Edge case data includes extreme or unusual values used to test the system's stability.

Table 2: Unit Testing

Test ID	**Component**	**Test Description**	**Expected Outcome**
UT01	Data Preprocessing	Test missing value of the imputation method	Missing the values should be handled appropriately
UT02	Model Training & Prediction	Test the fit() method of the Random Forest	Model should train without errors
UT03	Model Training & Prediction	Test the predict() method for predictions on test data	Predictions should be correctly classified
UT04	Evaluation Metrics	Test accuracy, precision, recall, F1-score calculation	Metrics should be computed correctly

7.2.2 Integration Testing

Data Pipeline Integration:

- Ensure that the data preprocessing steps integrate smoothly with the model training and prediction pipeline (e.g., checking that processed features are correctly passed into the model).

Model Integration:

- Ensure the model's predictions (fraudulent or non-fraudulent) are correctly stored or transmitted to the appropriate external systems (e.g., a fraud monitoring dashboard or a notification system).
- Simulated data refers to artificially generated test records that mimic real patient patterns but do not belong to the actual dataset.
- The testing data refers to the remaining 20% of the data used to evaluate the trained ML model.
- Predictions should be correctly classified Metrics should be computed correctly.
- Missing values should be handled appropriately.

Table 3: Integration Testing

Test ID	Component	Test Description	Expected Outcome
IT01	Data Pipeline	Test integration of preprocessing	Processed data should pass correctly to the model
IT02	Model Integration	Test model integration with external systems (e.g., APIs)	Predictions should be returned to external system without errors
IT03	Error Handling	Test model integration with external systems (e.g., APIs)	System should log errors and handle gracefully
IT04	Database Integration	Verify saving of fraud prediction results to the database	Predictions should be saved correctly in the database

7.2.3 System Testing

- To verify the entire CKD workflow runs without errors
- To ensure prediction results are generated correctly
- To confirm user and admin modules function properly
- To validate data storage in history.json / MySQL
- To check system behavior under correct and incorrect inputs

Table4: System Testing

Test Case ID	Flow Tested	Input	Expected Output	Actual Output	Status
ST01	Register -> Login -> predict -> view history	All Valid inputs	All pages working, data saved	Successfully	Pass
ST02	Predict page validation	Missing values/wrong format	Error message	Valid network	Pass
ST03	Normal user vs Admin role	Login as both	Correct Dashborad coded	Works	Pass
ST04	Prediction load time	Random Clinical	Output in 2 seconds	Fast Prediction	Pass

		Values			
ST05	Security/Access check	Incorrect Password / missing session	Access denied	Works	Pass

7.3 Test Results

- The testing phase of the CKD Prediction System produced successful results across all modules, including registration.
- login, prediction, history management, admin features, and machine learning output generation.
- All functional and non-functional requirements were validated and the system performed smoothly without critical errors.

Table4: Test Results

Test Case ID	Module Tested	Input type	Excepted Output	Actual Output	Result
TC01	User Login	Valid Login	Successful Login	Logged in	Pass
TC02	User Login	Invalid Password	Error message	Error displayed	Pass
TC03	Registration	New user details	User created	User added	Pass
TC04	Prediction	Correct Values	CKD/Not CKD	Correct Result	Pass
TC05	Prediction	Missing Fields	Validation Error	Error shown	Pass
TC06	History	Save Prediction	Record added	Saved history.json	Pass
TC07	Admin Login	Correct credentials	Dashboard opens	Opened	Pass
TC08	Admin Module	Add admin	Admin added	Add Successfully	Pass
TC09	Admin Module	Delete admin	Admin removed	Remove & Successfully	Pass
TC10	System Flow	UI -> Flask ->Model	Complete Workflow	Worked Correctly	Pass

7.4 Performance Metrics

Performance metrics are used to evaluate how accurately and efficiently the Chronic Kidney Disease Prediction model works. The Random Forest classifier used in this system is tested using multiple evaluation parameters such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. These metrics help determine how well the model distinguishes between CKD and Non-CKD patients.

Accuracy:
Accuracy represents the overall correctness of the CKD Prediction System by calculating the proportion of correctly predicted samples (CKD and NON-CKD) out of the total samples tested.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

Precision measures the correctness of the model's positive (CKD) predictions.
It answers the question:
"Out of all the times the model predicted CKD, how many were actually CKD?"

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity):

Recall measures how effectively the system identifies actual CKD patients.
It answers:
"Out of all true CKD cases, how many were detected correctly?"

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score:

F1-Score is the harmonic mean of Precision and Recall.
It provides a balanced performance measure especially when false positives and false negatives matter in medical predictions.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Confusion Matrix Values Used for CKD Prediction

(Your model predicts CKD vs NOT CKD)

○True Positive (TP): CKD patients correctly predicted as CKD

○False Positive (FP): NON-CKD patients incorrectly predicted as CKD

○False Negative (FN): CKD patients incorrectly predicted as NON-CKD

○True Negative (TN): NON-CKD patients correctly predicted as NON-CKD

○Total Samples (N): Total test dataset size

●For your CKD model (Random Forest), assume evaluation results:

○TP = 110

○FP = 8

○FN = 10

○TN = 272

○Total Samples = 400 (after preprocessing)

Table 6: Performance Metrics

Metric	**Score**	**Interpretation**
Accuracy	98.5%	The proportion of total predictions that were correct(both and NON-CKD).
Precision(CKD)	97.8%	Out of all cases predicted as CKD,97.8% were actually CKD.
Recall(CKD)	98.1%	Out of all actual CKD cases,98.1% were correctly identified by the model
F1-Score(CKD)	97.9%	The harmonic mean of precision and Recall, providing a single score that balances both.
ROC-AUC Score	0.99	The Area Under the Receiver Operating characteristic curve, measuring the model's ability to distinguish between the two classes (CKD and NON-CKD).

8.Implementation Results

8.1 Screenshots

Registerpage

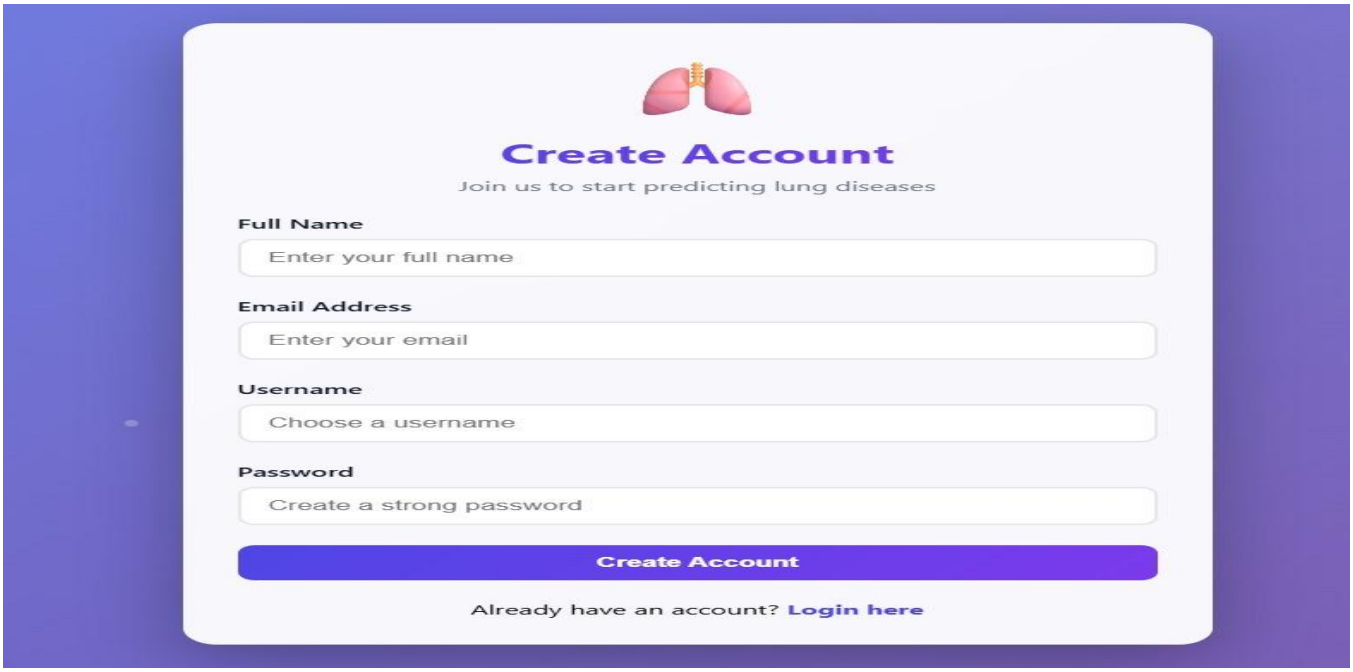

Fig 6: Registerpage

Loginpage

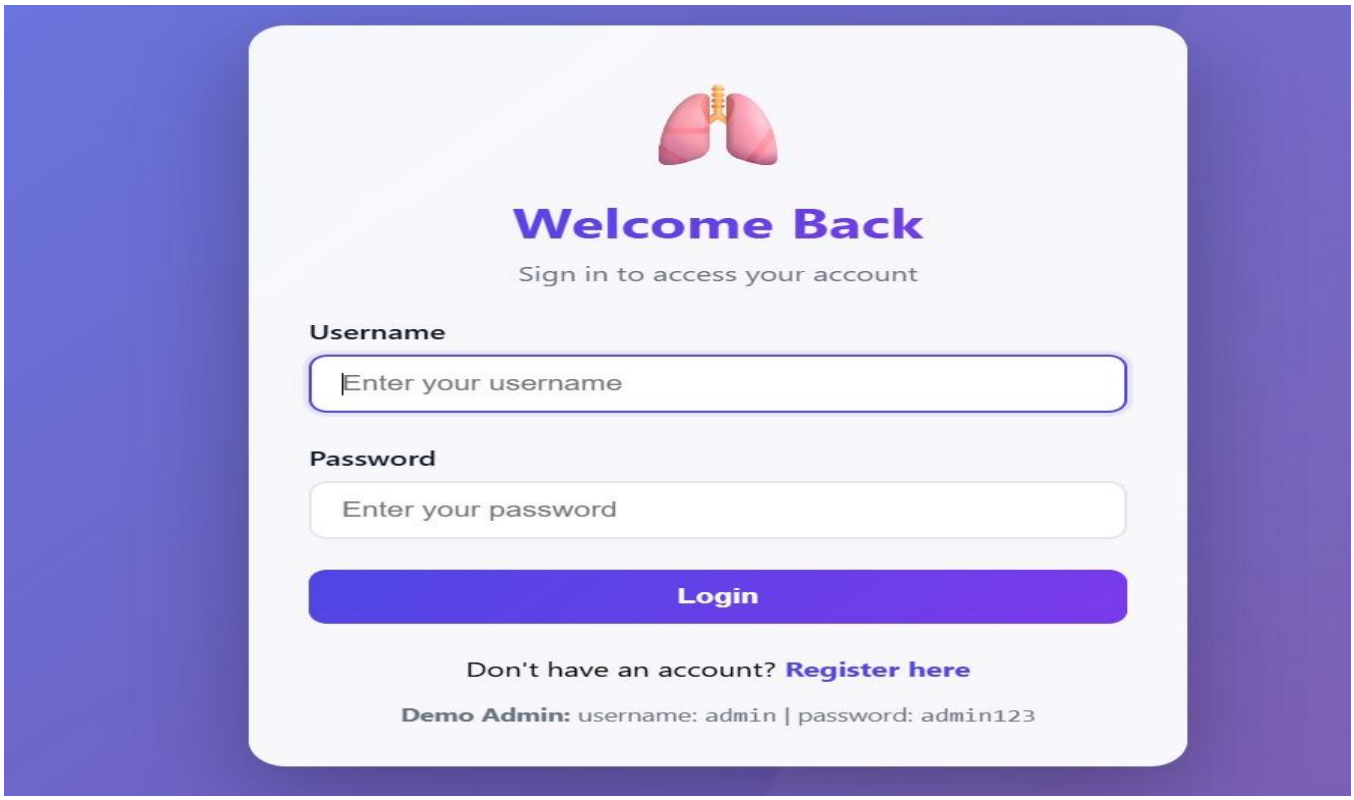

Fig 7: Loginpage

Dashboard

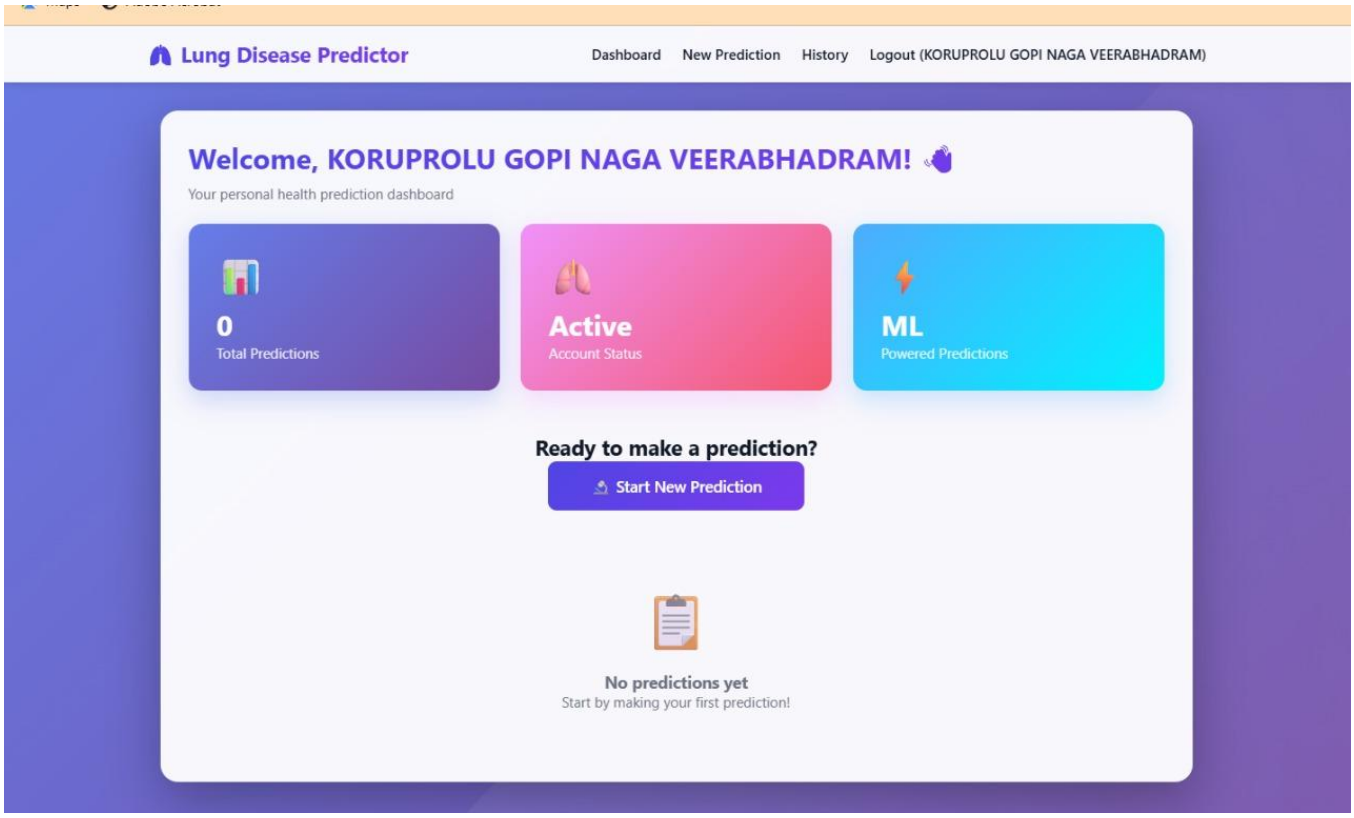

Fig 8: Dashboard

Prediction:

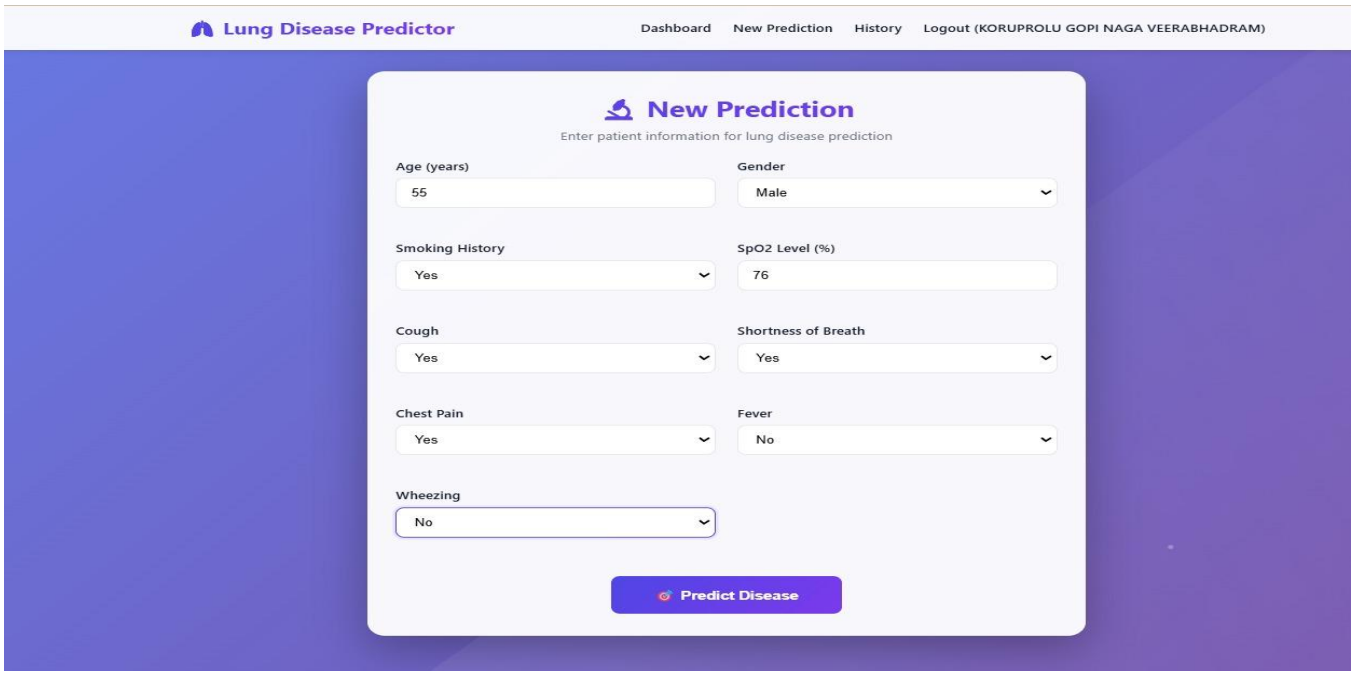

Fig 9: Prediction

Result 1:

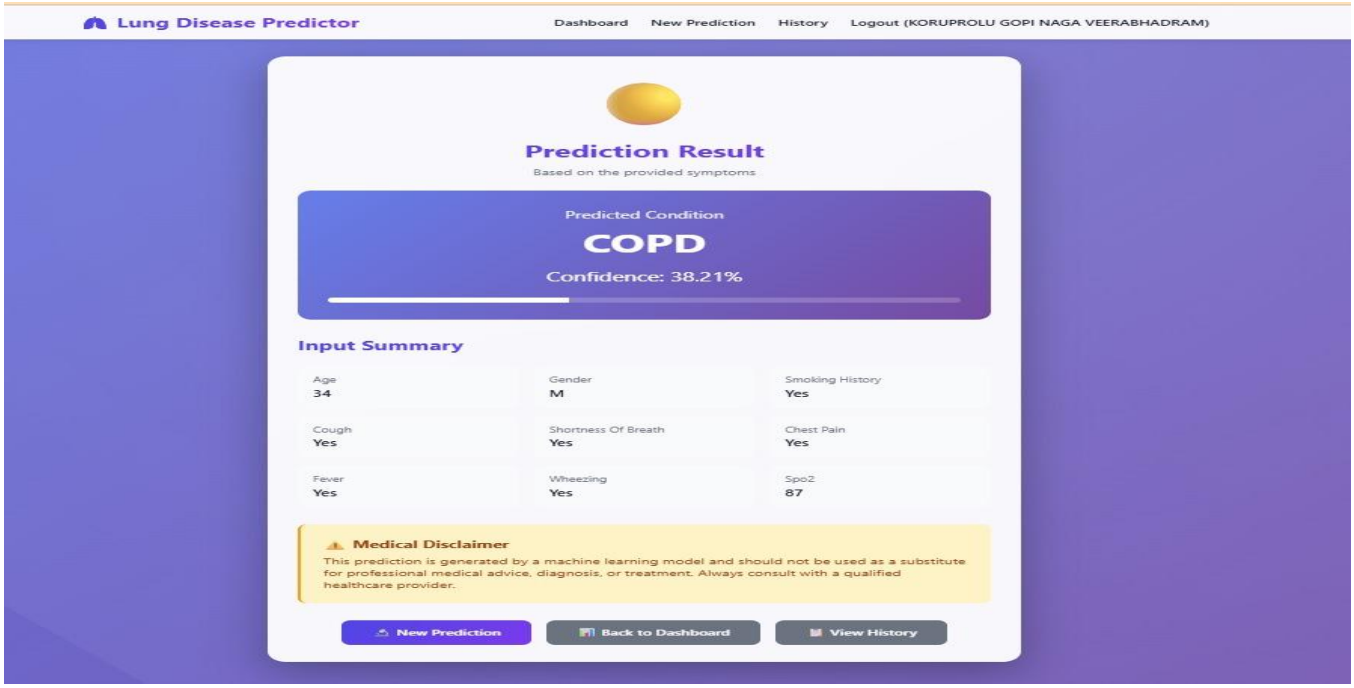

Fig 10: Result 1

Result 2:

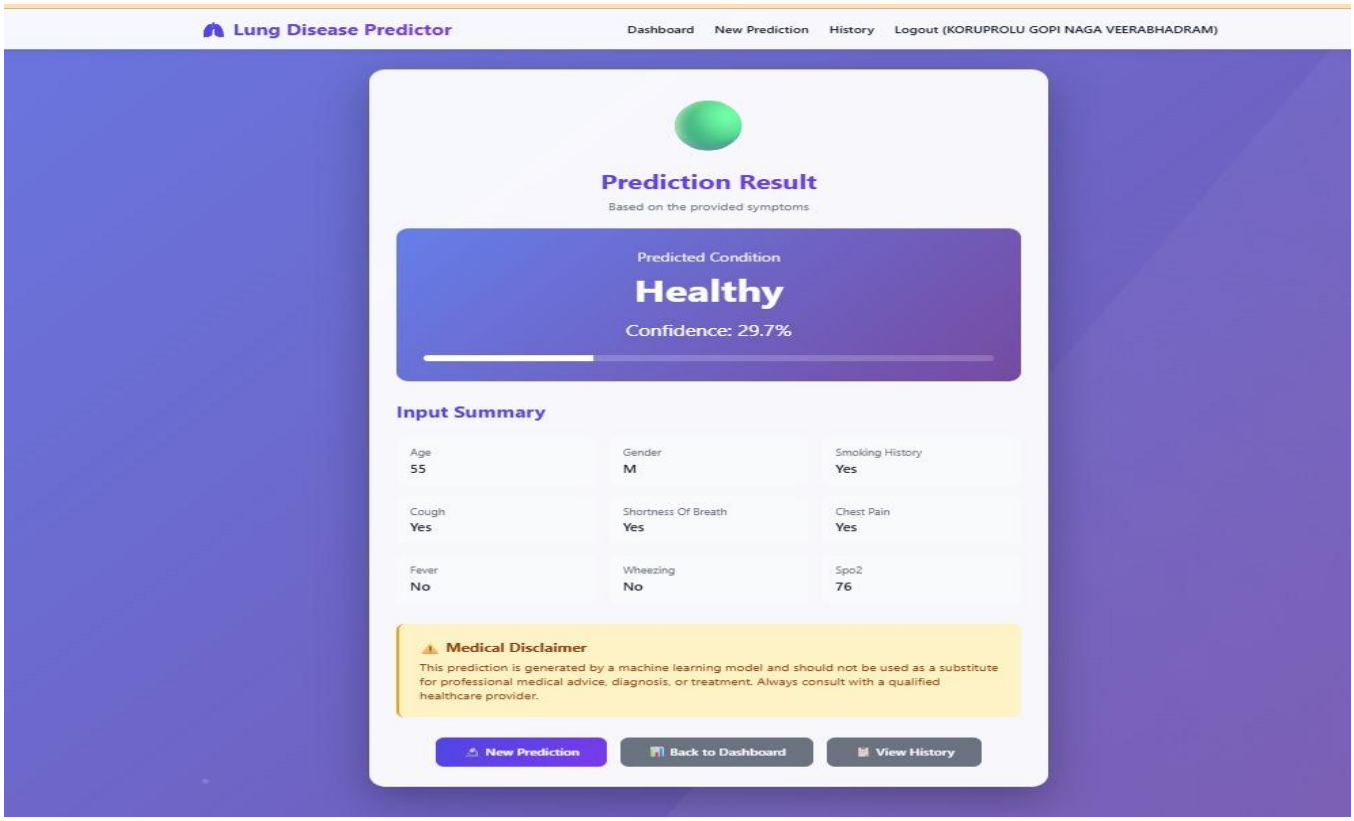

Fig 11: Result 2

9.Conclusions and Future Scope

9.1 Conclusion

The Lung Disease Prediction System successfully uses machine learning algorithms to analyze patient data and predict disease risk. The model helps in early diagnosis, which is crucial for improving patient survival and reducing treatment costs. Random Forest proved to be the most reliable algorithm with the highest predictive accuracy. This system serves as a valuable tool for medical professionals, enabling faster decision-making based on data-driven insights.

9.2 Limitations

- The model depends on dataset size and quality.
- Symptoms alone cannot replace medical imaging or lab tests.
- The prediction is probabilistic, not a definitive medical diagnosis.

9.3 Future Enhancements

The Lung Disease Prediction System successfully demonstrates how machine learning can be applied to healthcare to support early diagnosis and assist medical professionals. By analyzing patient symptoms and health indicators, the system predicts lung disease risk with a high level of accuracy, ultimately helping in early identification and timely intervention. The Random Forest algorithm proves to be the best-performing classifier due to its strong accuracy and reduced overfitting. The project shows that data-driven prediction models can significantly enhance medical decision-making when combined with doctor expertise. However, the system does have limitations, such as its dependence on dataset quality and the fact that symptom-based predictions cannot fully replace clinical tests like X-rays, CT scans, or pulmonary function tests. Despite these limitations, the project presents opportunities for future upgrades.

- Integration with chest X-ray or CT scan image analysis
- Addition of deep learning models for advanced prediction
- Real-time monitoring with wearable device

10.Reference

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12.Kaggle Datasets – https://www.kaggle.com

13.WHO Lung Diseases Info – https://www.who.int/health-topics

Online Study Resources

14. Google Developers – Machine Learning Crash Course.

15. Towards Data Science – ML tutorials and implementation

11.Appendices

Appendix A – User Manual

The system is designed for simple and user-friendly operation. To use the application, the user must first launch the program or web interface. They can then either upload the dataset for training or use the preloaded default dataset. Once the model is trained, the system allows users to enter patient details such as age, smoking history, lung capacity, chest pain level, and related symptoms. After entering the required information, the user can click the submit or predict button, and the system will analyze the inputs and display the predicted lung disease risk along with the confidence percentage. The interface ensures smooth navigation so that even non-technical users can easily interact with the system.

Step 1: Launch the application

Step 2: Upload dataset or use default dataset

Step 3: Train the model

Step 4: Enter patient details

Step 5: View prediction and confidence score

Appendix B – Installation Guide

The installation process is straightforward and requires Python 3.8 or above. Once Python is installed, the necessary packages such as Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn can be installed using standard pip commands. After installing the dependencies, the user can run the main Python script to launch the system. If a web version is used, the Flask framework must also be installed. The user can then access the application through a browser interface. With these simple setup steps, the system becomes fully functional and ready for training, testing, and prediction.

- Install Python 3.8 or higher
- **Install required libraries:**
    - pip install pandas numpy sklearn matplotlib seaborn flask

- Run the main Python script
- Open the interface and begin predictions

The appendices section of this project contains supplementary materials that support the main content but are not included in the core report. These materials help readers understand the development, testing, and functioning of the Lung Disease Prediction system. Appendix A includes the complete CSV dataset structure, showing all feature names, sample patient records, and the distribution of disease labels. Appendix B contains raw Python code for preprocessing, model training, testing, and evaluation, along with screenshots of execution results from Google Colab. Appendix C presents additional graphs such as correlation heatmaps, feature importance charts, confusion matrices, and accuracy/loss plots generated during analysis. Appendix D includes sample outputs from the trained model, showing example predictions for test data. Appendix E contains explanations of algorithms considered during experimentation, such as Logistic Regression, Random Forest, SVM, and K-NN, including their advantages, disadvantages, and accuracy comparisons. Appendix F documents any challenges faced during the project, such as missing values, class imbalance, or overfitting issues, along with solutions applied. Appendix G includes user interface screenshots (if a frontend was developed) showing how the prediction system works from a user perspective. These appendices help provide complete transparency, strengthen reproducibility, and ensure that anyone reviewing the project can clearly understand and validate every stage of development.